Syllabus
CPSC 431, Database and Applications – Spring 2026
Section 02

Description & Objectives

Database design and application development techniques for a real-world system. System analysis, requirement specifications, conceptual modeling, logic design, physical design, and web interface development. Develop projects using contemporary database management system and web-based application development platform.

Prerequisites

CPSC 332; Computer Science or Computer Engineering major or minor; or Computer Science or Computer Engineering graduate standing

Instructor

Professor Thomas L Bettens
Phone: (657) 278-2937
eMail: TBettens@fullerton.edu
Office: CS-209
Office Hours:
- Scheduled through Calendly at https://calendly.com/tbettens/office-hours
  - 15-minute meetings, in-person or virtual via Zoom
  - Check Calendly for available times. Only available slots for the next 7 days are shown.
  - Appointment suggested but not required. I'll hang out for a little while waiting for walk-ins. Just pop in the zoom room or office during posted (on Calendly) office hours. Happy to help if time permits.
- Special request appointments for both in-person and virtual meetings are available, time permitting, if coordinated at least 2 business days in advance
- By special request appointment only during final exam week

Meeting Information

Room: LH 401A*
Time: Tuesday & Thursday 4:00 PM - 5:15 PM
*In-Person course as defined in UPS 411.104. Note that up to 20%, or three of the 15 weeks, may be online. Course may include some synchronous or asynchronous online meetings/activities.

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Learning Goals

1. Students will learn the PHP fundamentals and lay the foundation for preparing themselves to become a professional PHP developer of enterprise web applications.

G.E. Requirements

This class does not meet any CSU General Education requirements.

Important Dates

CSUF's Academic and Admissions Calendars are posted online. The Academic Calendar contains all the campus closures and holidays you should be aware of. The Admissions Calendar contains all the major dates with respect to adding, dropping, and withdrawing from your classes. The final exam schedule is determined by Admissions and Records. Makeup exams are only available by advance request for documented exceptional circumstances.

- http://apps.fullerton.edu/AcademicCalendar/
- http://records.fullerton.edu/apps/calendars.aspx
- http://www.fullerton.edu/scheduling/final_exam_schedule

Monday, January 19	Martin Luther King, Jr. Day - No classes or office hours
Tuesday, January 20	First day of classes
Sunday, January 25	Last day students will be enrolled from waitlist
Tuesday, February 03	Last day to drop class without a grade of "W" *serious and compelling reasons* ***required*** *beyond this point (rare)*
Friday, February 13	Presidents' Day Observed - No classes or office hours
Thursday, March 19	Midterm Exam
Monday-Sunday, March 30-April 05	Spring Recess - No Classes or office hours
Friday, April 17	Last day to withdraw with a grade of "W"
Friday, May 08	Last day of classes
Tuesday, May 12	Final Exam 5:00PM - 6:50PM

Textbooks

Required

PHP and MySQL Web Development; Addison-Wesley Professional; 5th Edition, ISBN-13: 978-0321833891

PHP and MySQL Web Development, ISBN 9780321833891, is eligible to participate in the Titan Direct Access Program. This program offers increased student savings with immediate digital access to the material through the student portal. Students will enjoy free digital access to the content for the first week of classes. This option is only available in the student portal and students will receive email notifications when the content becomes available.

Titan Direct Access is an opt-out program, so if students prefer not to participate, they should simply opt-out of the materials through the link in the student portal. If students decide to maintain their access, they will be billed on their Titan Shops account after the free access period ends. Students will receive several email notifications of their need to make a payment. Those with outstanding balances 1 week after they've been billed will have their digital access and their bill removed with no further action needed.

Many popular technical books may be read online through the campus's subscription to Safari Books Online. The campus library's WWW proxy will grant you access from outside the campus network at http://www.library.fullerton.edu/asp/ipcheck.aspx?url=http://proquest.safaribooksonline.com/ ?uicode=calstate.

The Safari Books Online service can be accessed directly from any computer on the campus network at

http://proquest.safaribooksonline.com/.

The librarians at the CSUF Pollack Library have developed several research guides to support students studying computer science. The LibGuides are online at http://libraryguides.fullerton.edu/sb.php?subject_id=17830.

Development Tool Resources

Web application development

A suite of software applications and tools, collectively called the software stack, are required. We'll be using Apache, MariaDB, PHP, and Perl hosted on Ubuntu 24.04 in class. This set can be downloaded from XAMPP Installers and Downloads for Apache Friends and should be installed on your development machine. It is recommended that you use a machine separate from your mission critical machine (the one you depend on for your schoolwork) such as a virtual machine or a loaner laptop. We want to protect your mission critical machine from conflicting settings, potential security vulnerabilities, erasing or otherwise corrupting data, and so on.

No Use of AI Tools Permitted

This course assumes that work submitted for a grade by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as TitanGPT, ChatGPT, GitHub CoPilot, etc.). In this course, every element of class assignments must be fully prepared by the student. The use of generative AI tools for any part of your work will be treated as plagiarism. If you have questions, please contact me.

Academic Dishonesty

Students are encouraged to assist one another and discuss the course materials with your peers. It is your responsibility to be aware of and follow the spirit of CSU Fullerton's academic honesty policy which can be found at UPS 300.021 (https://www.fullerton.edu/senate/publications-policies-resolutions/ups/UPS%20300/UPS%20300.021.pdf). Academic dishonesty will not be tolerated. The University Catalog and the Class Schedule provide a detailed description of Academic Dishonesty under University Regulations.

By submitting work for evaluation, you acknowledge that you have adhered to the spirit of the university's academic honesty policy and that your submission is an original work by you unless otherwise directed to work in groups. Failure to follow the spirit of the academic honesty policy will result in a severely negative evaluation of the work in question and may result in involving the Department Chair and the Judicial Affairs office to seek a disciplinary remedy.

Course Outline

Week	Week Starting	Topic		Tentative Textbook Chapter
1	1/19	Introduction		
2	HW0 Due 1/26	**Part I:** Using PHP		Chapters 1 thru 7
3	2/2			
4	HW1 Due 2/9	**Part II:** Using MySQL		Chapters 8 thru 11
5	2/16			
6	2/23			
7	HW2 Due 3/2	**Part III:** Web Application Security		Chapters 14 thru 16
8	3/9			
9	HW3 Due 3/16			
		Midterm Exam (Thursday)		
10	3/23	**Part IV:** Advanced PHP Techniques • Using Session Control in PHP	**Final Project** (Chapter 29) • Kickoff	Chapter 22
	3/30	*Spring Recess, no class*		
11	4/6	• Implementing Authentication Methods	• Q&A open lab	Chapter 16
12	4/13	• Logging in and authenticating users • Managing password concepts	• Midway progress check • Open lab	Chapter 27
13	4/20	• Role Based Access	• Q&A and open lab	All
14	4/27	Final Project Oral Presentations		
15	5/4			
	5/11	Final Project's Product Delivery Due (Final Exam)		

Technical Proficiency

Technical proficiency in programming and software engineering should correspond to the prerequisite(s) of the course. Students are expected to be intimately familiar with their development platform of choice and be able to write and debug code in PHP and HTML at a level of proficiency that corresponds to the prerequisites of the course.

Technical proficiency with information technology, such as, but not limited to, the use of web-based online services, sending and receiving electronic mail, and desktop computer file systems, is assumed.

Grading

Plus and minus grading is not used when determining final grades.

Final grades are computed by first finding the average score in each category described in the table below on the right. All scores are normalized to a scale of 0 to 100 before being averaged. The average score for each category is then used to compute the weighted average according to the weights in the table below on the left.

Grade	% of Total Points
A	90–100%
B	80–89%
C	70–79%
D	60–69%
F	Below 59%

Category	% of Final Grade
Participation	5%
Assignments	20%
Project	20%
Midterm	25%
Final Exam	30%

Assignments

Programming and written assignments will be discussed in class and posted to the course website in advance of their due dates. Each assignment description will include the assignment's grading rubric. Reading assignments are outlined in the syllabus and it is the responsibility of the student to stay up to date with the reading.

Written assignments must be typeset and presented in a professional manner. Presentation, spelling, and grammar can be worth up to 30% of a written assignment's grade.

All programming assignments must be written in the HTML, PHP, and SQL programming languages, unless specified otherwise. Coding style must conform to professional norms. At a minimum, code must be commented, have descriptive names for identifiers, and contain a comment at the top of each file with pertinent information such as the student's name, email address, and assignment name. A plain text README.TXT must be included with each assignment submission summarizing and documenting the work submitted. For students unfamiliar with coding style, Google's style guides are an excellent starting point, https://github.com/google/styleguide, particularly their C++ style guide, https://google.github.io/styleguide/cppguide.html.

At the start of the semester, the instructor will detail the platform and tools used to grade student assignments. It is the student's responsibility to ensure that the assignments execute to his or her satisfaction on the instructor's grading platform.

There are approximately:

- 3 programming assignments
- 1 project cumulating with design artifacts and an oral presentation
- 1 midterm and 1 final exam
- 2 written assignments
- 14 weeks of reading assignments

Exceptions are made on a case-by-case basis given enough time and evidence to weigh the merits of the application.

Attendance Policy

Attending class is mandatory. Missing class is not allowed unless it is excused by the instructor. Missing class as part of a documented accommodation is guaranteed to be excused. The ADA accommodated student must make a reasonable effort to coordinate any absences with the instructor.

Make Up Policy

Exams and quizzes cannot be taken after they have been given in class. Due to an act of nature, personal medical emergency, a family crisis, an act of terrorism, severe civil unrest, etc. students have 10 calendar days to petition the instructor to retake any exam/quiz or submit an assignment without late penalty.

Exceptions shall be made on a case-by-case basis, provided there is time to evaluate the merits of such an application.

Participation

In the context of this course, participation is defined as the following:

- Arriving to class prepared and on time.
- Taking notes.
- Actively listening to the lecture and asking questions when appropriate.
- Annotating code listings and handouts.
- Bringing any required materials to class.
- When needed/desired, seeking assistance to complete assignments.
- Barring an emergency, not leaving the class session early unless the instructor consents.
- Not distracting oneself or others with smartphones, games, online diversions, etc.
- Respecting and treating the instructor and the student's peers civilly.

Required Material

- A writing instrument
- A notebook
- A USB memory stick
- A personal computer with the requisite development tools or regular access to a computer lab

Students with Special Needs

Please inform the instructor during the first week of classes about any disability or special needs that you may have that may require specific arrangements related to attending class sessions, carrying out class assignments, or writing papers or examinations. According to California State University policy, students with disabilities must document their disabilities at the Disability Support Services (DSS) Office in order to be accommodated in their courses. Additional information can be found at the DSS website, by calling 657-278-3112 or email dsservices@fullerton.edu.

Student Resources

Any student who wishes to discuss any concern may contact the assistant deans of the college. Assistant deans are student advocates who will help you navigate the university's policies and procedures and assist with resolving any conflicts.
Engineering & Computer Science - Dean's Office

Emergency Procedures

For your own safety and the safety of others, each student is expected to read and understand the guidelines published at Emergency Preparedness & Continuity (https://police.fullerton.edu/emergency-preparedness/). Should an emergency occur, follow the instructions given to you by faculty, staff, and public safety officials. An emergency information recording is available by calling the Campus Operation and Emergency Closure line at 657-278-4444.

Instructional Continuity

Due to an event such as an epidemic or a natural disaster that disrupts normal campus operations, students must monitor the course Canvas site and their campus email address for any instructions and assignments that the instructor announces.

Laboratory Safety

Safety is no accident. Learning and following the appropriate safety practices and protocols is an integral part to all laboratory courses. Following the appropriate safety practices and protocols minimizes the chances of repetitive stress injuries, mishandling hazardous materials, and injury to self and others. Additional campus laboratory safety information regarding hazardous materials is online at http://riskmanagement.fullerton.edu/laboratorysafety/.

Extra Credit

There are no planned opportunities for extra credit. If an opportunity arises it will be made available to everyone through Canvas

Recording & Transcription of Class Content

Recording class content is governed by UPS 330.230, UPS 330.230 (https://www.fullerton.edu/senate/publications-policies-resolutions/ups/UPS%20300/UPS%20330.230.pdf). Each instructor must permit class content to be recorded or transcribed by students when mandated to do so by the Americans with Disabilities Act or by other federal or state laws. Any recording of class content is for private use and study and shall not be made publicly accessible without the written consent of the instructor and students in the class.

Course Rules & Classroom Management

Unless an agreement or accommodation is reached between the student and the instructor, these rules must be followed.

- Attendance at all regularly scheduled lecture and discussion section is mandatory.
- Do not eat during lecture.
- If it makes noise, silence it.
- Computer use is not allowed in lecture except for taking notes.
- The student is responsible to be aware of any course announcements including changes to due dates and requirements.
- Homework, programming assignments, etc. may not be submitted late.
- Third party work (code, artwork, etc.) may not be used in student work without prior instructor consent. Failure to gain and document instructor consent will be construed as willful academic dishonesty.
- When a third party's work is incorporated into student work after gaining instructor consent, failure to wholly document the work's origin, copyright and license will be construed as willful academic dishonesty.